CURRICULUM VITAE

July 2026

Nakbin Choi, Ph.D.

Research Assistant Professor
Department of Civil, Urban, Earth, and Environmental Engineering
Ulsan National Institute of Science and Technology (UNIST)
50 UNIST-gil, Ulju-gun, Ulsan 44919, Republic of Korea
nbchoi21@unist.ac.kr
(Korea) 010-4946-6138 | (US) +1 (703) 843-4060

EDUCATION

2009.03 – 2015.02	**B.S. in Earth Science and Engineering**, UNIST, South Korea
2015.03 – 2021.02	**Ph.D. in Environmental Science and Engineering**, UNIST, South Korea **Dissertation:** *Development of a Coupled Data Assimilation System in the Fully Coupled Model and Its Implications for Seamless Prediction* **Advisor:** Prof. Myong-In Lee

EMPLOYMENTS

2023.06 – Present	**Postdoctoral Research Fellow**, George Mason University
2021.03 – 2023.06	**Postdoctoral Research Associate**, UNIST

RESEARCH EXPERIENCES

2023.01 – 2023.05	**Visiting Scientist**, Met Office Hadley Centre, UK
2017.02	**Visiting Researcher**, NASA GSFC, Climate and Radiation Lab, USA
2013.06 – 2013.08	**Internship**, NASA GSFC, GMAO, USA

PROFESSIONAL SERVICES

- Editor, Korean Atmospheric Scientists in America (KASA), 2024.10 – 2026.10

AWARDS

- Young Atmospheric Scientist Award, Korean Meteorological Society, 2025
- Outstanding Poster Presentation Award, Korean Meteorological Society, 2019

PATENT

- Medium-Range heatwave forecasting system and method. Domestic Patent Application, South Korea, 2019 (Myong-In Lee, Nakbin Choi, & Sunlae Tak)

PUBLICATIONS

[19] **Choi, N.**, Quan, X-W., and Stan, C.: Drivers of the Asymmetric ITCZ Bias in Climate Models, Scientific Reports (in Review), 2026.

[18] , 2026.

[17] Xavier, P., Willett, M., Graham, T., Earnshaw, P., Copsey, D., Narayan, N., Marzin, C., Sellar, A., Ackerley, D., Blaker, A., Blockley, E., Bodas-Salcedo, A., Bushell, A., **Choi, N.**, Chua, X. R., Guiavarc'h, C., Hassim, M., Heming, J., Hudson, D., Ineson, S., Jones, A., Jones, C., Keane, R., Kim, K., Kim, J., Kuhlbrodt, T., Lee, M.-I., Levine, R., Li, C., Martin, G., Megann, A., Mccabe, A., Moise, A., Regayre, L., Ridley, J., Roberts, L., Sahany, S., Schiemann, R. K. H., Storkey, D., Tennant, W., Tomassini, L., Tsushima, Y., Weedon, G. P., West, A., Wheeler, M. C., Williams, K., Zhou, X., and Zhu, H.: Assessment of the Met Office Global Coupled model version 5 (GC5) configurations, Geoscientific Model Development Discussion (in Review), 2026.

[16] **Choi, N.**, Stan, C., Lee, B., & Seo, E. Potential sources of predictability for the surface air temperature over the CONUS during boreal winter. Journal of Climate, e250495.

[15] **Choi, N.**, Stanczak, J., & Stan, C. (2026). The Role of Atmosphere–Ocean Coupling on the Prediction of Madden–Julian Oscillation. Journal of Climate, 39(13), 3669-3684.

[14] Kim, K., Lee, M. I., Lee, S., **Choi, N.**, Jin, F. F., Min, S. K., & Kam, J. Local and Remote Drivers of Increased Variability in Extreme Wildfire Conditions in Southeastern Australia. Available at SSRN 5708203.

[13] Lee, S., Lee, M. I., Lee, S., Jin, F. F., **Choi, N.**, & Tak, S. (2025). Distinct Features of the Summer Arctic Oscillation. Journal of Climate.

[12] **Choi, N.**, Lee, M. I., Ham, Y. G., Hyun, Y. K., Lee, J., & Boo, K. O. (2025). Reducing initialization shock by atmosphere-ocean coupled data assimilation and its impacts on the subseasonal prediction skill. Journal of Climate, 38(6), 1389-1401.

[11] **Choi, N.**, Lee, M. I., Tak, S., Cha, D. H., & Watanabe, M. (2025). Siberian heat extremes caused by Eurasian atmospheric teleconnections and amplified by local land surface conditions. Environmental Research Letters, 20(3), 034029.

[10] **Choi, N.**, & Stan, C. (2025). Large-Scale Surface Air Temperature Bias in Summer over the CONUS and Its Relationship to Tropical Central Pacific Convection in the UFS Prototype 8. Journal of Climate, 38(1), 117-129.

[9] Kim, H., **Choi, N.**, Lee, M. I., Tak, S., Cha, D. H., & Min, S. K. (2024). Causes on the 2018 record-breaking heatwave in South Korea: compound impacts from recurrent large-scale atmospheric teleconnections. Environmental Research Letters, 19(12), 124064.

[8] Tak, S., **Choi, N.**, Lee, J., & Lee, M. I. (2024). Probabilistic medium-range forecasts of extreme heat events over East Asia based on a global ensemble forecasting system. Weather and Climate Extremes, 45, 100694.

[7] **Choi, N.**, & Lee, S. (2023). Comparison of the opposite behaviours of Korean heatwaves with extreme hot sea surface temperatures in August 2016 and 2022. International Journal of Climatology, 43(15), 6993-7002.

[6] Lee, S., Park, M. S., Kwon, M., Park, Y. G., Kim, Y. H., & **Choi, N.** (2023). Rapidly changing East Asian marine heatwaves under a warming climate. Journal of Geophysical Research: Oceans, 128(6), e2023JC019761.

[5] Xavier, P., Willett, M., Graham, T., Earnshaw, P., Copsey, D., Narayan, N., Marzin, C., Zhu, H., Sellar, A., Ackerley, D., Blockley, E., Bodas-Salcedo, A., Bushell, A., **Choi, N.**, Chua, X. R., Guiavarc’h, C., Hassim, M., Heming, J., Hudson, D., Ineson, S. et al, (2023) Assessment of the Met Office Global Coupled model version 5 (GC5)configurations. Report. Met Office, Exeter. pp41.

[4] Yasunari, T. J., Nakamura, H., Kim, K. M., **Choi, N.**, Lee, M. I., Tachibana, Y., & da Silva, A. M. (2021). Relationship between circum-Arctic atmospheric wave patterns and large-scale wildfires in boreal summer. Environmental Research Letters, 16(6), 064009.

[3] **Choi, N.**, Lee, M. I., Cha, D. H., Lim, Y. K., & Kim, K. M. (2020). Decadal changes in the interannual variability of heat waves in East Asia caused by atmospheric teleconnection changes. Journal of Climate, 33(4), 1505-1522.

[2] **Choi, N.**, Kim, K. M., Lim, Y. K., & Lee, M. I. (2019). Decadal changes in the leading patterns of sea level pressure in the Arctic and their impacts on the sea ice variability in boreal summer. The Cryosphere, 13(11), 3007-3021.

[1] **Choi, N.**, & Lee, M. I. (2019). Spatial variability and long-term trend in the occurrence frequency of heatwave and tropical night in Korea. Asia-Pacific Journal of Atmospheric Sciences, 55, 101-114.